

DSO-Led Fair Dynamic Operating Envelopes for Aggregator-Level and Prosumer-Level Flexibility Coordination

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Abstract—

Index Terms—distributed energy resources, dynamic operating envelopes, feasible operating region, aggregators, distribution systems, fairness.

I. INTRODUCTION

THE growth of distributed energy resources (DERs), including rooftop photovoltaic (PV) systems and battery energy storage systems (BESSs), creates new opportunities for distribution networks to provide flexibility to the upstream grid [1]. Aggregators enable this participation by coordinating small-scale resources into portfolios for energy, ancillary services, and local flexibility markets [2], [3]. In distribution systems with multiple aggregators, flexibility provision also requires coordination among actors whose objectives need not coincide. Local prosumer operation, aggregator portfolio objectives, distribution system operator (DSO) security requirements, and upstream service requests must be reconciled. Prosumers operate their PV and BESS assets to serve their own consumption first. What a distribution network can offer upstream is therefore a residual: the capability that remains once local operation is committed. Aggregating device-level capabilities does not guarantee that the resulting portfolio flexibility can be delivered at the point of common coupling (PCC). The response available at the PCC depends on network operating conditions and, for BESSs, on the energy state established by previous operation. A network-level representation is therefore needed to assess the grid-feasible flexibility available at the PCC.

Region-based methods provide such a representation through the feasible operating region (FOR). The FOR describes the set of active-reactive power exchanges that a distribution system can deliver at the PCC without violating network or device constraints. Early studies established P - Q interface regions for transmission-distribution coordination [4], [5]. Random-sampling and OPF-based methods for computing these regions were compared and validated in [6]. Subsequent work examined how DER capabilities, integrated MV-LV constraints, and detailed network representations affect the size and shape of these regions [7], [8],

and robust methods have characterized the capability area under operational uncertainty [9]. More recent region-based formulations have considered unbalanced multiphase networks [10] and time-varying operating regions at feeder and end-user connection points [11]. These formulations capture how flexibility changes with network and operating conditions. When BESSs are included, operating periods are also linked because charging and discharging decisions modify the state of charge and change the flexibility available later.

This dependence has motivated multi-period and online approaches to flexibility characterization. Aggregation and fair disaggregation of DER flexibility were formalized with zonotopic sets in [12]. Chen *et al.* represented feasible aggregate power trajectories with disaggregation guarantees and incorporated them into a model-predictive control framework [13]. Chen and Li developed multi-period active-power intervals and P - Q regions with disaggregation guarantees [14], while Wen *et al.* characterized and optimized time-coupled flexibility under distribution-network security constraints [15], [16]. At the TSO-DSO interface, López *et al.* developed multi-period flexibility regions conditioned on deployed flexibility [17], and online methods have used dispatch feedback to update future portfolio flexibility [18]. These approaches capture important links between present operation and future flexibility. The setting considered here introduces a specific operational sequence: local operation is first established through a network-constrained self-consumption dispatch, and the BESS actions implemented in that dispatch determine the state from which the next PCC FOR is constructed. In this way, upstream flexibility is a residual quantity. It is evaluated from the state the local dispatch leaves behind, not from an independently assumed BESS state.

A dispatch-conditioned PCC FOR determines the flexibility that remains available to the upstream grid, but it does not determine how a selected response should be realized by the actors behind the PCC. Operating-envelope methods provide time-varying active-power limits for DER owners and aggregators [19], [20], with extensions to P - Q operating regions [11], [21] and allocation schemes based on proportional, technical, and equity-based criteria at the prosumer level [22]–[24]. These allocation schemes respond to earlier evidence that network location alone creates unequal treatment among prosumers [25], [26] and that reactive-power control can reduce this inequality [27]. For systems with multiple aggregators, network-secure operating regions and coordina-

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TABLE I
COMPARISON WITH THE CLOSEST REGION- AND ENVELOPE-BASED
METHODS

	P - Q PCC region	Dispatch- cond. rolling	Multiple aggreg.	Two-level fairness	Unbal. MV-LV	Activation guarantee
Chen <i>et al.</i> [13]	•	○	—	—	○	•
Lankeshwara <i>et al.</i> [11]	•	○	—	—	○	•
Russell <i>et al.</i> [28]	•	—	○	—	○	•
Jiang and Guo [30]	—	—	•	○	—	•
Gao <i>et al.</i> [24]	○	—	—	○	○	•
Lukač <i>et al.</i> [32]	○	—	—	○	○	•
This paper	•	•	•	•	•	•

• addressed ○ partial — not considered

tion mechanisms have also been developed to manage network access and shared constraints [28]–[30], and fairness has been considered in time-dependent P - Q flexibility provision [31]. Reactive-power flexibility has also been used to improve the fairness of active-power export allocations, although at the prosumer level only [32]. These methods primarily determine permissible participant operation, network access, or flexibility allocation. The activation problem considered here starts from a different decision point: an upstream service request has selected a specific operating point from the PCC FOR, and coordinated participant setpoints must reproduce that exchange while respecting the distribution network. With multiple aggregators, this realization also introduces two allocation levels, first among aggregators and then among the prosumers within each aggregator. Fairness at both levels is an operational requirement rather than a design preference. A DSO-led scheme should treat competing aggregators without discrimination, and repeated activations should not concentrate effort on prosumers at electrically favorable network locations.

The reviewed formulations therefore address complementary parts of the flexibility-provision problem, but they do not jointly establish the operational chain considered in this work. Table I positions the proposed framework against the closest region- and envelope-based methods. In particular, the local dispatch that determines the current BESS state must be linked to the flexibility offered in the next interval, and a selected point from that offer must be linked to its network-feasible realization across multiple aggregators and prosumers. An operationally consistent PCC offer should be network-feasible when constructed, reachable from the state established by the implemented dispatch, and realizable by the participating resources when activated. Without these links, an offered region may be feasible for an assumed operating state but unavailable after the actual BESS operation, or a valid PCC request may lack a corresponding network-feasible and fair participant-level dispatch.

Accordingly, this paper develops a DSO-led rolling-horizon framework that connects network-constrained self-consumption, dispatch-conditioned PCC flexibility, and two-level fair activation across multiple aggregators and their prosumers. At each rolling step, a network-constrained self-consumption problem establishes local operation and determines the BESS actions implemented in the current interval.

The resulting state of charge is carried forward to construct the next PCC FOR. Once an upstream service request selects an operating point from this region, the required deviation from the self-consumption baseline is coordinated first among aggregators and then among their prosumers while preserving network feasibility. The main contributions are as follows:

- 1) A rolling-horizon construction that quantifies the grid-feasible P - Q flexibility remaining at the PCC after prosumer self-consumption is committed. The network-constrained dispatch fixes the BESS state from which the next-interval region is built, so every offered operating point is reachable from implemented operation under the adopted three-phase MV-LV formulation.
- 2) A network-feasible hierarchical activation formulation is developed for a selected PCC operating point, with fairness enforced at two levels: first among competing aggregators and then among the prosumers within each aggregator. The formulation coordinates the required deviation from the self-consumption baseline while preserving the requested PCC exchange and distribution-network constraints.
- 3) A comprehensive operational assessment quantifies the effects of dispatch commitment and two-level fairness on reachable flexibility, resource utilization, BESS throughput, and computational performance. The analysis also distinguishes the complete feeder-level PCC flexibility, which reflects the combined effect of demand, PV generation, controllable resources, and network conditions, from the controllable BESS contribution around the established distribution-system operating point.

The remainder of this paper is organized as follows. Section II presents the proposed rolling-horizon framework, including state propagation, the self-consumption and reachable-region formulations, and the hierarchical activation method. Section ?? describes the unbalanced MV-LV test system, DER data, fairness criteria, and simulation setup. Section ?? evaluates the reachable P - Q regions, service activation, resource allocation, and computational performance. Section ?? summarizes the main findings and concludes the paper.

II. ROLLING-HORIZON COORDINATION FRAMEWORK FOR REACHABLE FORS

The proposed framework coordinates the flexibility of prosumer-owned PV and BESS units. Every 30 minutes, the DSO computes the feasible operating region (FOR) of active and reactive power at the distribution network PCC. Each operating point within the FOR reflects the self-consumption dispatch implemented by the prosumers and is therefore operationally achievable.

In this work, *self-consumption* denotes the use of local PV and BESS resources to meet household demand before providing system flexibility. *Flexibility* denotes the residual active and reactive power capability available at the PCC after this dispatch is committed. A *rolling-horizon* scheme links both stages by re-optimizing a 48-slot window every half-hour.

At each rolling step, the methodology solves two optimization problems in sequence. Problem 1 (P1) computes a

network-constrained self-consumption dispatch and commits the current BESS actions. The batteries discharge during PV deficits and charge during PV surpluses, minimizing prosumer–network power exchanges and determining the battery states entering the next interval.

Problem 2 (P2) fixes the BESS actions committed by P1 and applies a directional optimal power flow (OPF) to compute the FOR for the next interval. The 48 half-hourly FORs describe what the DN can deliver at the PCC throughout the day. The complete process is summarized in Fig. 1 and detailed below.

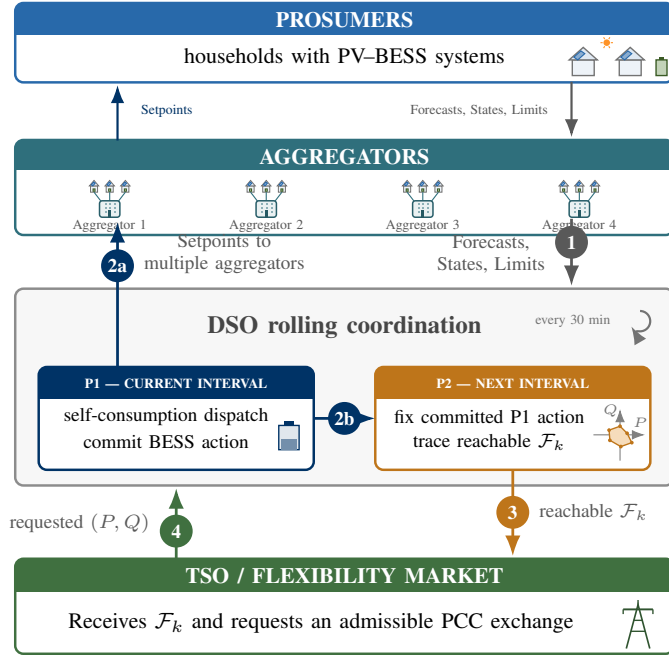


Fig. 1. Rolling-horizon coordination among prosumers, aggregators, the DSO, and the TSO. Aggregators are the sole interface between prosumers and the DSO. P1 commits the current BESS action (Steps 1–2a), P2 conditions the next-interval FOR on that action (Step 2b), and the TSO requests an exchange within the offered region (Steps 3–4).

The DSO coordinates the process based on its network model, measurements, and OPF capability [19], [20], [33]. In Step 1, aggregators convey prosumer forecasts, measured BESS states, and operating limits to the DSO. After solving P1, the DSO sends the first-interval BESS setpoints through the aggregators for implementation (Step 2a).

The BESS actions committed by P1 are fixed in all P2 directional solves (Step 2b), ensuring that the FOR boundary is computed from a common post-dispatch state. The DSO communicates the resulting region $\mathcal{F}_k$ to the TSO or flexibility market (Step 3). Any exchange $(P, Q) \in \mathcal{F}_k$ may be requested (Step 4), as described in Section XX.

Both problems are built on a common prosumer, BESS, and three-phase network model adapted from the convex multi-period OPF in [8]. Their distinction lies in the objective functions and variables fixed in each formulation. The following subsections detail the rolling-horizon timing, common model, P1 and P2 formulations, and DN-scale implementation.

A. Rolling-Horizon Timing and State Propagation

At each rolling step k , the DSO optimizes a 24-h window of $H = 48$ half-hour slots, indexed by $h \in \mathcal{H} = \{1, \dots, 48\}$. Slot $h = 1$ represents the current interval, and slot $h = 2$ represents the next interval.

The physical periods of the day are indexed by $t \in \{1, \dots, 48\}$. At step k , these periods are ordered as

$$(k, k+1, \dots, 48, 1, \dots, k-1),$$

with the physical period corresponding to slot h given by

$$t_k(h) = 1 + \text{mod}(k + h - 2, 48). \quad (1)$$

The first two slots couple the self-consumption dispatch and the FOR computation. Slot $h = 1$ is the *operation slot*, in which the BESS actions obtained from P1 are committed and implemented. Slot $h = t^* = 2$ is the *flexibility slot*, for which P2 computes the FOR while retaining the operation-slot actions. Slots $h \geq 3$ provide the look-ahead required to preserve intertemporal and network feasibility, as illustrated in Figure. 2.

Although the full window is optimized, only the slot-1 BESS charging and discharging commands, $\hat{P}_{s,1|k}^{b,+}$ and $\hat{P}_{s,1|k}^{b,-}$, are implemented. These commands determine the BESS energy entering the flexibility slot:

$$\hat{E}_{s,1|k}^b = E_{s,0|k}^b + \Delta t \left(\eta_s^{b,+} \hat{P}_{s,1|k}^{b,+} - \frac{\hat{P}_{s,1|k}^{b,-}}{\eta_s^{b,-}} \right), \quad \forall s \in \mathcal{S}, \quad (2)$$

where $\Delta t = 0.5$ h.

The committed commands are fixed in every directional solve of P2. Thus, all FOR boundary points are evaluated from the same post-dispatch energy state $\hat{E}_{s,1|k}^b$, ensuring reachability from the implemented P1 dispatch.

When the horizon advances, the flexibility slot of step k becomes the operation slot of step $k+1$, since

$$t_k(2) = t_{k+1}(1).$$

Thus, $\hat{E}_{s,1|k}^b$ is the energy predicted by the model at the beginning of the next rolling step. After the slot-1 commands are implemented, this prediction is replaced by the measured BESS energy:

$$E_{s,0|k+1}^b = E_{s,0|k+1}^{b,\text{meas}}(t_k(2)), \quad \forall s \in \mathcal{S}. \quad (3)$$

This measurement-based update closes the rolling-horizon loop and accounts for deviations between the predicted and realized BESS states.

Repeating this procedure throughout the day produces the sequence

$$\{\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_{48}\}, \quad (4)$$

where $\mathcal{F}_k$ denotes the FOR computed at step k for physical period $t_k(2)$.

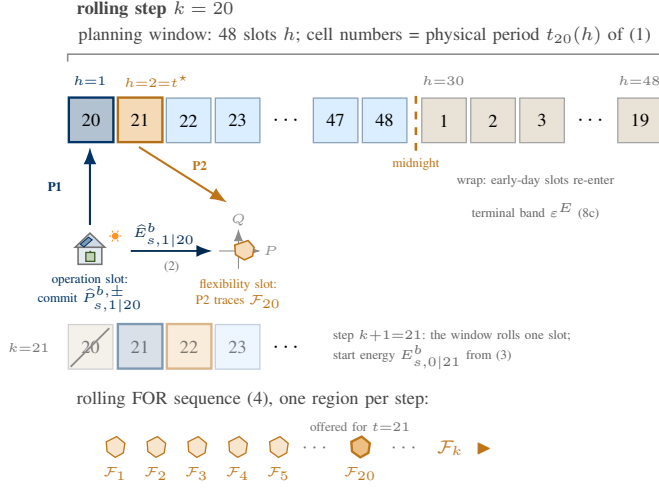


Fig. 2. Rolling-horizon coupling of P1 and P2 at step $k = 20$. P1 commits the BESS action for the operation slot ($h = 1$, period 20); P2 fixes that action and traces $\mathcal{F}_{20}$ for the flexibility slot ($h = t^* = 2$, period 21). Later slots keep the committed action consistent over the wrapped day, and the executed energy initializes step $k + 1$.

1) *Prosumer and BESS Model*: For each prosumer $s \in \mathcal{S}$ and slot $h \in \mathcal{H}$, $P_{s,h}^d$ and $P_{s,h}^g$ denote the forecast demand and PV generation, respectively. Both are assumed known, as forecast uncertainty is outside the scope of this work.

The net active-power exchange with the distribution network is

$$P_{s,h} = P_{s,h}^d - P_{s,h}^g + P_{s,h}^b, \quad \forall s \in \mathcal{S}, h \in \mathcal{H}, \quad (5)$$

where $P_{s,h}^b$ denotes the net BESS power. Positive values indicate network import and BESS charging, respectively.

Following [8], the grid and BESS powers are decomposed into nonnegative directional variables:

$$P_{s,h} = P_{s,h}^+ - P_{s,h}^-, \quad P_{s,h}^+, P_{s,h}^- \geq 0, \quad (6a)$$

$$P_{s,h}^b = P_{s,h}^{b,+} - P_{s,h}^{b,-}, \quad P_{s,h}^{b,+}, P_{s,h}^{b,-} \geq 0. \quad (6b)$$

Here, $P_{s,h}^+$ and $P_{s,h}^-$ denote network import and export, while $P_{s,h}^{b,+}$ and $P_{s,h}^{b,-}$ denote BESS charging and discharging. This continuous representation avoids binary operating-mode variables and preserves convexity. Fig. 3 illustrates this operation.

Let $E_{s,0}^b$ denote the BESS energy at the beginning of the optimization window. The energy at the end of slot h is

$$E_{s,h}^b = E_{s,0}^b + \Delta t \sum_{\tau=1}^h \left(\eta_s^{b,+} P_{s,\tau}^{b,+} - \frac{P_{s,\tau}^{b,-}}{\eta_s^{b,-}} \right), \quad (7)$$

where $\eta_s^{b,+}$ and $\eta_s^{b,-}$ are the charging and discharging efficiencies, respectively, and $\Delta t = 0.5$ h.

The BESS energy, power, and terminal-state limits are

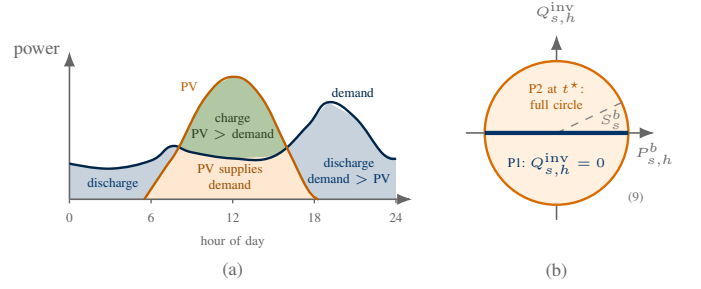


Fig. 3. BESS operation in the two optimization problems. (a) Typical prosumer demand $P_{s,h}^d$ and PV generation $P_{s,h}^g$ over one day. Under self-consumption, PV generation supplies the local demand, the BESS charges during a PV surplus, and discharges during a PV deficit, reducing the residual exchange with the network. (b) Inverter capability defined by (9). P1 operates along the $Q_{s,h}^{inv} = 0$ diameter, whereas P2 may use the complete P - Q capability during the flexibility slot $h = t^*$.

$$\underline{E}_s^b \leq E_{s,h}^b \leq \overline{E}_s^b, \quad (8a)$$

$$0 \leq P_{s,h}^{b,+}, P_{s,h}^{b,-} \leq \overline{P}_s^b, \quad (8b)$$

$$-\varepsilon^E \overline{E}_s^b \leq E_{s,H}^b - E_{s,0}^b \leq \varepsilon^E \overline{E}_s^b. \quad (8c)$$

where (8c) limits the net energy deviation over the window and prevents end-of-horizon depletion.

The BESS inverter capability is constrained by

$$(P_{s,h}^b)^2 + (Q_{s,h}^{inv})^2 \leq (S_s^b)^2, \quad \forall s \in \mathcal{S}, h \in \mathcal{H}, \quad (9)$$

where $Q_{s,h}^{inv}$ denotes the inverter reactive power and S_s^b its apparent-power rating.

2) *Three-Phase Network Model*: The network constraints are adopted from the convexified three-phase AC OPF formulation developed in [8], [34] and are not reproduced in full here.

In brief, the model represents phase voltages and currents in rectangular coordinates, includes mutual coupling between phases, and accounts for the fixed tap Dy11 transformer. Constant-power injections and the minimum-voltage constraint are linearized around a reference voltage profile that incorporates the MV-LV phase displacement.

The BESS decisions are coupled to the network currents through

$$P_{s,h}^b = V_{n,\phi,h}^{\text{re}} I_{s,h}^{b,\text{re}} + V_{n,\phi,h}^{\text{im},0} I_{s,h}^{b,\text{im}}, \quad (10a)$$

$$Q_{s,h}^{inv} = V_{n,\phi,h}^{\text{im},0} I_{s,h}^{b,\text{re}} - V_{n,\phi,h}^{\text{re},0} I_{s,h}^{b,\text{im}}. \quad (10b)$$

The PCC exchange used in the P2 objective is

$$P_h^{\text{PCC}} = \sum_{\phi \in \Phi} (V_{n_0,\phi}^{\text{re}} I_{n_0,\phi,h}^{\text{re}} + V_{n_0,\phi}^{\text{im}} I_{n_0,\phi,h}^{\text{im}}), \quad (11a)$$

$$Q_h^{\text{PCC}} = \sum_{\phi \in \Phi} (V_{n_0,\phi}^{\text{im}} I_{n_0,\phi,h}^{\text{re}} - V_{n_0,\phi}^{\text{re}} I_{n_0,\phi,h}^{\text{im}}), \quad (11b)$$

where n_0 denotes the fixed voltage PCC bus and positive P_h^{PCC} denotes import. The reachable FOR is constructed from these exchanges at $h = t^*$.

The OPF enforces the branch-current and voltage limits

$$(I_{e,\phi,h}^{\text{re}})^2 + (I_{e,\phi,h}^{\text{im}})^2 \leq \bar{I}_e^2, \quad (12a)$$

$$(V_{n,\phi,h}^{\text{re}})^2 + (V_{n,\phi,h}^{\text{im}})^2 \leq \bar{V}_n^2, \quad (12b)$$

$$2V_{n,\phi,h}^{\text{re},0} V_{n,\phi,h}^{\text{re}} + 2V_{n,\phi,h}^{\text{im},0} V_{n,\phi,h}^{\text{im}} - |V_{n,\phi,h}^0|^2 \geq \underline{V}_n^2. \quad (12c)$$

““latex

B. Network-Constrained Self-Consumption Dispatch (P1)

P1 computes a network-feasible self-consumption schedule and determines the BESS action implemented at the current slot $h = 1$. Local PV generation supplies demand, surplus generation may charge the BESS, and stored energy may be discharged to reduce grid imports.

Over the common feasible set $\mathcal{X}_k$, P1 minimizes prosumer grid exchange and BESS throughput:

$$J^{\text{sc}} = \sum_{s \in \mathcal{S}} \sum_{h \in \mathcal{H} \setminus \{t^*\}} \omega_h^{\text{self}} (P_{s,h}^+ + P_{s,h}^-) + \mu \sum_{s \in \mathcal{S}} \sum_{h \in \mathcal{H}} (P_{s,h}^{b,+} + P_{s,h}^{b,-}). \quad (13)$$

The first term promotes self-consumption. Since

$$P_{s,h} = P_{s,h}^+ - P_{s,h}^-, \quad P_{s,h}^+, P_{s,h}^- \geq 0, \quad (14)$$

minimizing $P_{s,h}^+ + P_{s,h}^-$ reduces $|P_{s,h}|$, and therefore both imports and exports whenever they can be offset by local PV generation and BESS operation.

The flexibility slot $h = t^*$ is excluded from this term to avoid biasing the PCC exchange explored by P2. All device, energy, and network constraints remain enforced at this slot.

The second term penalizes BESS throughput, discouraging unnecessary cycling and simultaneous charging and discharging while preserving a continuous convex formulation. The coefficient μ is selected to make loss-driven charge–discharge cycles unattractive and is reported in Section ??.

At rolling step k , P1 is formulated as

$$(P1) \quad \min_{\mathbf{x}_k} J^{\text{sc}}, \quad (15a)$$

$$\text{s.t. } \mathbf{x}_k \in \mathcal{X}_k, \quad (15b)$$

$$Q_{s,h}^{\text{inv}} = 0, \quad \forall s \in \mathcal{S}, h \in \mathcal{H}. \quad (15c)$$

Constraint (15c) disables BESS reactive-power support during the baseline dispatch.

Only the current-slot setpoints

$$(\hat{P}_{s,1|k}^{b,+}, \hat{P}_{s,1|k}^{b,-}) \quad (16)$$

are implemented. The decisions for $h \geq 2$ ensure a feasible continuation over the remaining horizon. P2 fixes the implemented action and computes the reachable PCC exchanges at $h = t^*$.

C. Next-Interval Reachable FOR Construction (P2)

P2 constructs the network-feasible PCC exchanges available at the target slot $h = t^*$, conditioned on the current-slot BESS action committed by P1. The directional search set is

$$\Theta = \left\{ \theta_i = \frac{2\pi i}{N_\theta} \mid i = 0, \dots, N_\theta - 1 \right\}. \quad (17)$$

For each $\theta_i \in \Theta$, P2 solves

$$(P2) \quad \max_{\mathbf{x}_{k,i}} P_{t^*|k,i}^{\text{PCC}} \cos \theta_i + Q_{t^*|k,i}^{\text{PCC}} \sin \theta_i, \quad (18a)$$

$$\text{s.t. } \mathbf{x}_{k,i} \in \mathcal{X}_k, \quad (18b)$$

$$P_{s,1|k,i}^{b,+} = \hat{P}_{s,1|k}^{b,+}, \quad \forall s \in \mathcal{S}, \quad (18c)$$

$$P_{s,1|k,i}^{b,-} = \hat{P}_{s,1|k}^{b,-}, \quad \forall s \in \mathcal{S}, \quad (18d)$$

$$Q_{s,h|k,i}^{\text{inv}} = 0, \quad \forall s \in \mathcal{S}, h \in \mathcal{H} \setminus \{t^*\}. \quad (18e)$$

The objective traces a PCC boundary point in the direction θ_i . Constraints (18c) and (18d) fix the P1 action, ensuring that all directional solves enter the target slot from the same post-dispatch BESS state. The Constraint (18e) restricts the reactive power support to $h = t^*$, while all constraints in $\mathcal{X}_k$ remain enforced throughout the horizon. Figure 4 illustrates this construction.

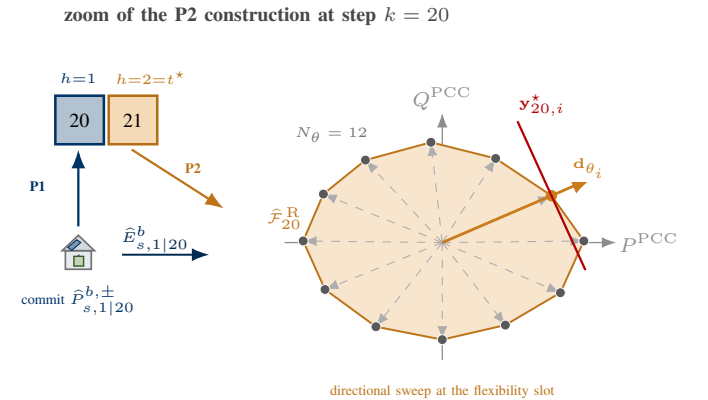


Fig. 4. P2 construction at rolling step $k = 20$. P1 commits the operation-slot BESS powers $\hat{P}_{s,1|20}^{b,\pm}$, determining the energy state $\hat{E}_{s,1|20}^b$ entering the flexibility slot $h = t^*$. P2 solves $N_\theta = 12$ directional OPFs, and the convex hull of the resulting boundary points $\mathbf{y}_{20,i}^*$ defines the reachable FOR $\hat{\mathcal{F}}_{20}^{\text{R}}$.

The optimal PCC exchange obtained in direction θ_i is

$$\mathbf{y}_{k,i}^* = \begin{bmatrix} P_{t^*|k,i}^{\text{PCC},*} \\ Q_{t^*|k,i}^{\text{PCC},*} \end{bmatrix}, \quad (19)$$

and the polygonal reachable FOR is

$$\hat{\mathcal{F}}_k^{\text{R}} = \text{conv} \{ \mathbf{y}_{k,i}^* \mid i = 0, \dots, N_\theta - 1 \}. \quad (20)$$

Each boundary point is supported by a feasible full-horizon trajectory consistent with the P1 action. Moreover, because $\mathcal{X}_k$ is convex, the commitment constraints are affine, and the PCC exchange is linear, the interior of $\hat{\mathcal{F}}_k^{\text{R}}$ is also reachable. For finite N_θ , the polygon provides an inner approximation of the complete reachable FOR.

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REFERENCES

- [1] International Energy Agency, "Unlocking the potential of distributed energy resources: Power system opportunities and best practices," IEA, Paris, France, Tech. Rep., 2022. [Online]. Available: <https://www.iea.org/reports/unlocking-the-potential-of-distributed-energy-resources>
- [2] A. M. Carreiro, H. M. Jorge, and C. H. Antunes, "Energy management systems aggregators: A literature survey," *Renewable and Sustainable Energy Reviews*, vol. 73, pp. 1160–1172, 2017. [Online]. Available: <http://dx.doi.org/10.1016/j.rser.2017.01.179>
- [3] M. S. Miah, R. Shah, M. U. Aslam, N. Amjadi, S. Walker, V. Terzija, S. Dehghan, and N. Mithulananthan, "Providing ancillary services by aggregated distributed energy resources of active distribution systems for the upstream grid: A critical review," *Renewable and Sustainable Energy Reviews*, vol. 233, p. 116858, 2026. [Online]. Available: <http://dx.doi.org/10.1016/j.rser.2026.116858>
- [4] F. Capitanescu, "TSO–DSO interaction: Active distribution network power chart for TSO ancillary services provision," *Electric Power Systems Research*, vol. 163, pp. 226–230, Oct. 2018. [Online]. Available: <http://dx.doi.org/10.1016/j.epr.2018.06.009>
- [5] J. Silva, J. Sumaili, R. J. Bessa, L. Seca, M. A. Matos, V. Miranda, M. Caujolle, B. Goncer, and M. Sebastian-Viana, "Estimating the active and reactive power flexibility area at the TSO–DSO interface," *IEEE Transactions on Power Systems*, vol. 33, no. 5, pp. 4741–4750, 2018. [Online]. Available: <http://dx.doi.org/10.1109/TPWRS.2018.2805765>
- [6] D. A. Contreras and K. Rudion, "Computing the feasible operating region of active distribution networks: Comparison and validation of random sampling and optimal power flow based methods," *IET Generation, Transmission & Distribution*, vol. 15, no. 10, pp. 1600–1612, Jan. 2021. [Online]. Available: <http://dx.doi.org/10.1049/gtd2.12120>
- [7] S. Riaz and P. Mancarella, "Modelling and characterisation of flexibility from distributed energy resources," *IEEE Transactions on Power Systems*, vol. 37, no. 1, pp. 38–50, Jan. 2022. [Online]. Available: <http://dx.doi.org/10.1109/TPWRS.2021.3096971>
- [8] L. Gutierrez-Lagos, K. Petrou, and L. F. Ochoa, "Quantifying the effects of medium voltage–low voltage distribution network constraints and distributed energy resource reactive power capabilities on aggregators," *IET Generation, Transmission & Distribution*, vol. 15, no. 14, pp. 2019–2032, Mar. 2021. [Online]. Available: <http://dx.doi.org/10.1049/gtd2.12152>
- [9] M. Kalantar-Neyestanaki, F. Sossan, M. Bozorg, and R. Cherkaoui, "Characterizing the reserve provision capability area of active distribution networks: A linear robust optimization method," *IEEE Transactions on Smart Grid*, vol. 11, no. 3, pp. 2464–2475, May 2020. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2019.2956152>
- [10] M. Prasad, Z. H. Rather, R. Razzaghi, and S. Doolla, "A new approach to determine feasible operating region of unbalanced distribution networks with distributed photovoltaics," *IEEE Transactions on Power Delivery*, vol. 40, no. 3, pp. 1493–1504, 2025. [Online]. Available: <http://dx.doi.org/10.1109/TPWRD.2025.3555210>
- [11] G. Lankeshwara, R. Sharma, R. Yan, T. K. Saha, and J. V. Milanović, "Time-varying operating regions of end-users and feeders in low-voltage distribution networks," *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 4600–4611, Mar. 2024. [Online]. Available: <http://dx.doi.org/10.1109/TPWRS.2023.3302421>
- [12] F. L. Muller, J. Szabo, O. Sundstrom, and J. Lygeros, "Aggregation and disaggregation of energetic flexibility from distributed energy resources," *IEEE Transactions on Smart Grid*, vol. 10, no. 2, pp. 1205–1214, Mar. 2019. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2017.2761439>
- [13] X. Chen, E. Dall'Anese, C. Zhao, and N. Li, "Aggregate power flexibility in unbalanced distribution systems," *IEEE Transactions on Smart Grid*, vol. 11, no. 1, pp. 258–269, Jan. 2020. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2019.2920991>
- [14] X. Chen and N. Li, "Leveraging two-stage adaptive robust optimization for power flexibility aggregation," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 3954–3965, 2021. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2021.3068341>
- [15] Y. Wen, Z. Hu, and L. Liu, "Aggregate temporally coupled power flexibility of DERs considering distribution system security constraints," *IEEE Transactions on Power Systems*, vol. 38, no. 4, pp. 3884–3896, 2023. [Online]. Available: <http://dx.doi.org/10.1109/TPWRS.2022.3196708>
- [16] Y. Wen, Y. Guo, Z. Hu, and G. Hug, "Quantifying and optimizing the time-coupled flexibilities at the distribution-level for TSO–DSO coordination," *IEEE Transactions on Power Systems*, vol. 40, no. 6, pp. 5071–5085, Nov. 2025. [Online]. Available: <http://dx.doi.org/10.1109/TPWRS.2025.3563041>
- [17] L. Lopez, A. Gonzalez-Castellanos, and D. Pozo, "Construction of multi-period TSO–DSO flexibility regions," in *2022 International Conference on Smart Energy Systems and Technologies (SEST)*. IEEE, 2022, pp. 1–6. [Online]. Available: <http://dx.doi.org/10.1109/SEST53650.2022.9898423>
- [18] X. Wang, S. Wang, L. Du, and Y. Chen, "Online hierarchical aggregate power flexibility characterization of EV charging stations with disaggregation feasibility guarantee," *IEEE Transactions on Transportation Electrification*, vol. 11, no. 6, pp. 13 274–13 286, Dec. 2025. [Online]. Available: <http://dx.doi.org/10.1109/TTE.2025.3593904>
- [19] K. Petrou, A. T. Procopiou, L. Gutierrez-Lagos, M. Z. Liu, L. F. Ochoa, T. Langstaff, and J. M. Theunissen, "Ensuring distribution network integrity using dynamic operating limits for prosumers," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 3877–3888, 2021. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2021.3081371>
- [20] M. Z. Liu, L. F. Ochoa, P. K. C. Wong, and J. Theunissen, "Using OPF-based operating envelopes to facilitate residential DER services," *IEEE Transactions on Smart Grid*, vol. 13, no. 6, pp. 4494–4504, Nov. 2022. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2022.3188927>
- [21] Y. Zabihinia Gerdroodbari, M. Khorasany, and R. Razzaghi, "Dynamic PQ operating envelopes for prosumers in distribution networks," *Applied Energy*, vol. 325, p. 119757, Nov. 2022. [Online]. Available: <http://dx.doi.org/10.1016/j.apenergy.2022.119757>
- [22] K. Petrou, M. Z. Liu, A. T. Procopiou, L. F. Ochoa, J. Theunissen, and J. Harding, "Operating envelopes for prosumers in LV networks: A weighted proportional fairness approach," in *2020 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*. IEEE, Oct. 2020, pp. 579–583. [Online]. Available: <http://dx.doi.org/10.1109/ISGT-Europe47291.2020.9248975>
- [23] M. R. Alam, P. T. H. Nguyen, L. Naranpanawe, T. K. Saha, and G. Lankeshwara, "Allocation of dynamic operating envelopes in distribution networks: Technical and equitable perspectives," *IEEE Transactions on Sustainable Energy*, vol. 15, no. 1, pp. 173–186, Jan. 2024. [Online]. Available: <http://dx.doi.org/10.1109/TSTE.2023.3275082>
- [24] Y. Gao, X. Xu, Z. Yan, and M. Shahidehpour, "Equitable active-reactive power envelopes for distributed energy resources in power distribution systems," *IEEE Transactions on Smart Grid*, vol. 16, no. 2, pp. 1480–1494, Mar. 2025. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2024.3524759>
- [25] R. Tonkoski, L. A. C. Lopes, and T. H. M. El-Fouly, "Coordinated active power curtailment of grid connected PV inverters for overvoltage prevention," *IEEE Transactions on Sustainable Energy*, vol. 2, no. 2, pp. 139–147, Apr. 2011. [Online]. Available: <http://dx.doi.org/10.1109/TSTE.2010.2098483>
- [26] M. Z. Liu, A. T. Procopiou, K. Petrou, L. F. Ochoa, T. Langstaff, J. Harding, and J. Theunissen, "On the fairness of PV curtailment schemes in residential distribution networks," *IEEE Transactions on Smart Grid*, vol. 11, no. 5, pp. 4502–4512, 2020. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2020.2983771>
- [27] D. Gebbran, S. Mhanna, Y. Ma, A. C. Chapman, and G. Verbič, "Fair coordination of distributed energy resources with Volt-Var control and PV curtailment," *Applied Energy*, vol. 286, p. 116546, Mar. 2021. [Online]. Available: <http://dx.doi.org/10.1016/j.apenergy.2021.116546>
- [28] J. S. Russell, P. Scott, and J. Iria, "Network-secure aggregator operating regions with flexible dispatch envelopes in unbalanced systems," *Electric Power Systems Research*, vol. 235, p. 110728, Oct. 2024. [Online]. Available: <http://dx.doi.org/10.1016/j.epr.2024.110728>
- [29] C. Chen, S. Bose, T. D. Mount, and L. Tong, "Wholesale market participation of DERs: DSO–DERA–ISO coordination," *IEEE Transactions on Power Systems*, vol. 39, no. 5, pp. 6605–6614, 2024. [Online]. Available: <http://dx.doi.org/10.1109/TPWRS.2024.3352003>
- [30] Z. Jiang and Y. Guo, "Bargaining-based approach for dynamic operating envelope allocation in distribution networks," *IEEE Transactions on Smart Grid*, vol. 16, no. 5, pp. 3588–3600, 2025. [Online]. Available: <http://dx.doi.org/10.1109/TSG.2025.3566419>
- [31] G. C. Kryonidis and C. S. Demoulias, "Fair provision of time-dependent flexibility at TSO–DSO interface," in *2025 IEEE Power & Energy Society General Meeting (PESGM)*. IEEE, 2025, pp. 1–5. [Online]. Available: <http://dx.doi.org/10.1109/PESGM52009.2025.11224998>
- [32] L. Lukač, T. Antić, and T. Capuder, "On the value of reactive power flexibility for fair export allocation under operating envelopes,"

- Sustainable Energy, Grids and Networks*, vol. 46, p. 102283, 2026. [Online]. Available: <http://dx.doi.org/10.1016/j.segan.2026.102283>
- [33] H. Früh, S. Müller, D. Contreras, K. Rudion, A. von Haken, and B. Surmann, "Coordinated vertical provision of flexibility from distribution systems," *IEEE Transactions on Power Systems*, vol. 38, no. 2, pp. 1834–1844, Mar. 2023. [Online]. Available: <http://dx.doi.org/10.1109/TPWRS.2022.3162041>
- [34] L. Gutierrez-Lagos, M. Z. Liu, and L. F. Ochoa, "Implementable three-phase OPF formulations for MV-LV distribution networks: MILP and MIQCP," in *2019 IEEE PES Innovative Smart Grid Technologies Conference - Latin America (ISGT Latin America)*. IEEE, 2019, pp. 1–6. [Online]. Available: <http://dx.doi.org/10.1109/ISGT-LA.2019.8895008>

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